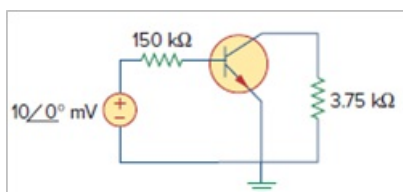


Problem

For the transistor amplifier of Fig. 19.60, find the voltage gain, current gain, input impedance, and output impedance. Assume that

$$h_{ie} = 6 \text{ k}\Omega, \quad h_{re} = 1.5 \times 10^{-4}, \quad h_{fe} = 200, \quad h_{oe} = 8 \text{ }\mu\text{S}$$

Figure 19.60



Step-by-step solution

Step 1 of 8

Refer to Figure 19.60 in the text book.

Calculate the voltage gain A_v of the transistor.

$$A_v = \frac{-h_{fe}R_L}{h_{ie} + (h_{ie}h_{oe} - h_{re}h_{fe})R_L}$$

Substitute $6 \text{ k}\Omega$ for h_{ie} , $8 \text{ }\mu\text{S}$ for h_{oe} , 1.5×10^{-4} for h_{re} , 200 for h_{fe} and $3.75 \text{ k}\Omega$ for R_L .

$$\begin{aligned} A_v &= \frac{-(200)(3.75 \text{ k})}{6 \text{ k} + ((6 \text{ k})(8 \text{ }\mu) - (1.5 \times 10^{-4})(200))(3.75 \text{ k})} \\ &= \frac{-750000}{6 \text{ k} + (0.048 - 0.03)(3.75 \text{ k})} \\ &= \frac{-750000}{6 \text{ k} + (0.018)(3.75 \text{ k})} \\ &= \frac{-750000}{6 \text{ k} + 67.5} \\ A_v &= \frac{-750000}{6067.5} \\ &= -123.61 \end{aligned}$$

Therefore, the voltage gain of the transistor is $\boxed{-123.61}$.

Step 2 of 8

A_v represents the ratio $\frac{V_o}{V_b}$.

$$A_v = \frac{V_o}{V_b}$$

Substitute -123.61 for A_v .

$$-123.61 = \frac{V_o}{V_b}$$

$$V_b = \frac{V_o}{-123.61}$$

Therefore, the expression for the base voltage is $\frac{V_o}{-123.61}$.

Step 3 of 8

Consider the mesh equation at the base circuit of the amplifier.

$$-V_s + I_b R_s + V_b = 0$$

$$V_s = I_b R_s + V_b$$

Substitute $\frac{h_{oe} V_o}{\frac{h_{fe}}{1 + h_{oe} R_L} - h_{fe}}$ for I_b and $\frac{V_o}{-123.61}$ for V_b .

$$\begin{aligned} V_s &= \frac{h_{oe} V_o R_s}{\frac{h_{fe}}{1 + h_{oe} R_L} - h_{fe}} + \left(\frac{V_o}{-123.61} \right) \\ &= \frac{h_{oe} R_s V_o}{\frac{h_{fe}}{1 + h_{oe} R_L} - h_{fe}} - \frac{V_o}{123.61} \end{aligned}$$

Substitute $8 \mu\text{S}$ for h_{oe} , 200 for h_{fe} , $150 \text{ k}\Omega$ for R_s and $3.75 \text{ k}\Omega$ for R_L .

$$\begin{aligned} V_s &= \frac{(8 \mu)(150 \text{ k}) V_o}{\frac{200}{1 + (8 \mu)(3.75 \text{ k}\Omega)} - 200} - \frac{V_o}{123.61} \\ &= \frac{1.2 V_o}{\frac{200}{1 + 0.03} - 200} - \frac{V_o}{123.61} \\ &= \frac{1.2 V_o}{\frac{200}{1.03} - 200} - \frac{V_o}{123.61} \\ &= \frac{1.2 V_o}{194.18 - 200} - \frac{V_o}{123.61} \end{aligned}$$

Step 4 of 8

On further simplification:

$$\begin{aligned} V_s &= \frac{1.2 V_o}{-5.82} - \frac{V_o}{123.61} \\ &= \left(-\frac{1.2}{5.82} - \frac{1}{123.61} \right) V_o \\ &= (-0.206 - 0.0081) V_o \\ &= (-0.2141) V_o \end{aligned}$$

$$\begin{aligned} \frac{V_o}{V_s} &= -\frac{1}{0.2141} \\ &= -4.671 \end{aligned}$$

$\frac{V_o}{V_s}$ represents the overall voltage gain.

Therefore, the voltage gain of the amplifier is -4.671.

Step 5 of 8

Calculate the current gain A_i of the transistor.

$$A_i = \frac{h_{fe}}{1 + h_{oe}R_L}$$

Substitute $8\ \mu\text{S}$ for h_{oe} , 200 for h_{fe} and $3.75\ \text{k}\Omega$ for R_L .

$$\begin{aligned} A_i &= \frac{200}{1 + (8\ \mu)(3.75\ \text{k})} \\ &= \frac{200}{1 + 0.03} \\ &= \frac{200}{1.03} \\ &= 194.17 \end{aligned}$$

Therefore, the value of the current gain is 194.17.

Step 6 of 8

Calculate the value of the input impedance of the transistor.

$$Z_{in} = h_{ie} - \frac{h_{re}h_{fe}R_L}{1 + h_{oe}R_L}$$

Substitute $6\ \text{k}\Omega$ for h_{ie} , $8\ \mu\text{S}$ for h_{oe} , 1.5×10^{-4} for h_{re} , 200 for h_{fe} and $3.75\ \text{k}\Omega$ for R_L .

$$\begin{aligned} Z_{in} &= 6\ \text{k} - \frac{(1.5 \times 10^{-4})(200)(3.75\ \text{k})}{1 + (8\ \mu)(3.75\ \text{k})} \\ &= 6\ \text{k} - \frac{112.5}{1 + 0.03} \\ &= 6\ \text{k} - \frac{112.5}{1.03} \\ &= 6\ \text{k} - 109.22 \\ Z_{in} &= 5.89\ \text{k} \\ &\approx 6\ \text{k}\Omega \end{aligned}$$

Therefore, the value of the input resistance of the transistor is 6 kΩ.

Step 7 of 8

Calculate the input resistance of the amplifier.

$$Z_{in,amp} = Z_{in} + R_s$$

Substitute $6\ \text{k}\Omega$ for Z_{in} and $150\ \text{k}\Omega$ for R_s .

$$\begin{aligned} Z_{in,amp} &= 6\ \text{k} + 150\ \text{k} \\ &= 156\ \text{k}\Omega \end{aligned}$$

Therefore, the value of the input resistance of the amplifier is 156 kΩ.

Step 8 of 8

Calculate the output resistance if the amplifier.

$$Z_{out} = \frac{R_s + h_{ie}}{(R_s + h_{ie})h_{oe} - h_{re}h_{fe}}$$

Substitute $6 \text{ k}\Omega$ for h_{ie} , $8 \text{ }\mu\text{S}$ for h_{oe} , 1.5×10^{-4} for h_{re} , 200 for h_{fe} , $150 \text{ k}\Omega$ for R_s and $3.75 \text{ k}\Omega$ for R_L .

$$\begin{aligned} Z_{out} &= \frac{150 \text{ k} + 6 \text{ k}}{(150 \text{ k} + 6 \text{ k})(80 \text{ }\mu) - (1.5 \times 10^{-4})(200)} \\ &= \frac{156 \text{ k}}{(156 \text{ k})(80 \text{ }\mu) - 0.03} \\ &= \frac{156 \text{ k}}{12.48 - 0.03} \\ &= \frac{156 \text{ k}}{12.45} \end{aligned}$$

$$Z_{out} = 125.30 \text{ k}\Omega$$

Therefore, the value of the output impedance is $125.30 \text{ k}\Omega$.